RAJALAKSHMI ENGINEERING COLLEGE

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T2

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|---|---|
| **Started on** | Thursday, 8 October 2026, 11:47 AM |
| **State** | Finished |
| **Completed on** | Thursday, 8 October 2026, 12:06 PM |
| **Time taken** | 19 mins 29 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

**For example:**

| Input | Result |
|---|---|
| 3<br>1 3 5<br>4 | 1 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n, k;
    scanf("%d", &n);

    int A[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &A[i]);

    scanf("%d", &k);

    int i = 0, j = 1;

    while (j < n) {
        int diff = A[j] - A[i];

        if (diff == k && i != j) {
            printf("1");
            return 0;
        }
        else if (diff < k) {
            j++;
        }
        else {
            i++;
        }

        if (i == j)
            j++;
    }

    printf("0");

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1 3 5<br>4 | 1 | 1 | ✔ |
| ✔ | 10<br>1 4 6 8 12 14 15 20 21 25<br>1 | 1 | 1 | ✔ |
| ✔ | 10<br>1 2 3 5 11 14 16 24 28 29<br>0 | 0 | 0 | ✔ |
| ✔ | 10<br>0 2 3 7 13 14 15 20 24 25<br>10 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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